Khusel Avirmed

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EDUCATION

Cornell University, College of Agriculture and Life Sciences **Graduation May 2021**
Bachelor of Science, *GPA 3.61*
Major 1: Environment and Sustainability, Concentration in Environmental Economics
Major 2: International Agriculture and Rural Development, Concentration in Development and Economics

HONORS/AWARDS

- Distinction in Research 2021
- Dean's List 2019-2021
- Sustainability Hackathon 2020
- Societal Solutions Scholar 2019 – Cornell SC Johnson College of Business

PROFESSIONAL EXPERIENCE

Research Assistant, *Cornell University* | Ithaca, NY **Sep 2020-Present**
Under the supervision of Prof. Chris B. Barrett

- Coded Python script to collect Mongolian rangeland quality geospatial data using Google Earth Engine (GEE)
- Coded R scripts to clean the resulting data set (a million rows) and performed quality assurance processing
- Assisted in designing and performing two stages Least Squares Regression analysis using Instrumental Variable to estimate the effect of climate change and livestock herd size on rangeland quality
- Conducted literature reviews and kept a record of memos synthesizing important findings from relevant scientific papers
- Designed and drafted manuscripts in Overleaf for science journals

Independent Project, *GreenSummer (*green-summer.herokuapp.com*)* | Ithaca, NY **June 2022-Present**
An open source, interactive digital platform monitoring rangeland quality of Mongolia in a near real time

- Developed production and development pipelines in Github to streamline geospatial data sets in GEE
- Coded Python script to streamline publicly available satellite images
- Coded a predictive machine learning model to analyze satellite images to produce an accurate rangeland quality indicator
- Coded spatial visualization to display the resulting indicator on the interactive web map

Co-Founder, *Agcess* (www.carbonetech.com) | Ithaca, NY **Oct 2020-Jan 2022**

- Created an easy access precision agriculture platform for small holder farmers to assist their sustainable decision making
- Wrote Python script to streamline high-resolution satellite images data processing in GEE
- Conducted customer discovery and collaborated with team to prototype the first minimum viable product

Remote Research Development Intern, *EnerGaia* | Bangkok, Thailand **Jun 2020 – Aug 2020**
CALS Global Fellows Program

- Developed high level plan to determine biomass density of spirulina growing in an urban environment using computer vision
- Researched the most recent patents and technologies used in the digital agriculture sector to determine the biomass of plants
- Curated potential options and methods such as machine learning, hyperspectral cameras, and color palette comparisons
- Deployed a minimal viable product using hyperspectral cameras to deliver a cost-effective solution

LEADERSHIP EXPERIENCE

Teaching Assistant, *Perspectives in Int. Development IARD 6960* | Ithaca, NY **Mar 2019-May 2021**

- Led Teaching Team with technical support for an online classroom with 50 students
- Took weekly attendance, organized monthly homework assignments, and responded timely to e-mail inquiries from students

Teaching Assistant, Indigenous *Ways of Knowing NTRES 3330* | Ithaca, NY **Aug 2020 - Dec 2020**

- Led sections under Prof. Karim-Aly Kassam for 46 students for hybrid classroom
- Graded weekly homework assignments as well as the mid-term projects and responded timely to e-mail inquiries from students
- Organized and oversee technical logistics of the hybrid classroom settings, including Zoom invitations, breakout rooms, online polls, and screen sharing while complying with University COVID rules and recommendations

Executive Board & Director of Education, *Social Enterprise at Cornell* | Ithaca, NY **Sep 2019 – Sep 2020**

- Led Education portfolio for a newly founded club of 60 members, providing educational and growth opportunities for members of an organization and of the campus community regarding environmental and social stewardship amid business worlds
- Founded inaugural 'Hub-Chats', a semester-long project to engage full-time social entrepreneurs and connect them with students in Cornell's community, with the primary goal to inspire conversation about social entrepreneurship on campus

SKILLS, ACTIVITIES & INTERESTS

Technical Skills: R, STATA (Familiar), Google Earth Engine, Github, ArcMap, Latex (Familiar)
Activities: Social Enterprise at Cornell, Beta Theta Pi, Cornell Academy Soccer Club, Mongolian Student Association at Cornell
Interests: Soccer, Basketball, Horse Riding, Hiking, and Reading